```
stream = spark.readStream \
  .format("cloudFiles") \
  .option("cloudFiles.format", "json") \
```

Figure 3.8 – Creating the stream using stream format CloudFiles and file format JSON

2. The default is to set column types to `string`. However, we can provide schema hints so that the columns we are sure about get typed appropriately. While inferring the schema, we also want to infer the column types. The new column is a `string`, so we do not see this option in action as new columns default to a `string`.

```
.option("cloudFiles.schemaHints","CustomerID int, Amount double,
TransactionTimestamp timestamp") \
.option("cloudFiles.inferColumnTypes","true") \
```

Figure 3.9 – Setting schema hints to reduce type mismatches; we want to infer the data type for columns that are not in the schema hint

3. Auto Loader uses the rescue column to catch the change and quickly puts the new column into play without data loss! Note that the stream will fail and need to be restarted. The schema location is needed because we want Auto Loader to keep track of the schema and evolve it over time.

```
.option("cloudFiles.schemaEvolutionMode", "addNewColumns") \
.option("cloudFiles.schemaLocation", schema_location) \
.load(raw_data_location) \
.select("*") \
```

Figure 3.10 – Setting schema evolution to add new columns and designating the schema location so Auto Loader keeps track of schema changes over time

4. Next, we specify the location of the raw data files. We need to select the fields we want. We select all data fields, see *Figure 3.10*, but you could be selective on the fields you pull in or add the metadata column, see File metadata column in Further reading for more information.
5. The code in *Figure 3.11* corresponds to the "write" portion of the stream. Delta is the default file format, but we prefer to set it explicitly for clarity. We also designate that this stream is append-only, as we are not yet performing any updates or inserts.
6. A checkpoint location is important. A checkpoint is a mechanism in Spark Structured Streaming that allows you to save the state of your stream. This is where it keeps track of its place in the processing of files. If your stream fails, when it restarts, it uses the checkpoint to resume processing where it left off. The merge schema option is essential for schema evolution. Merging the schema adds the new columns as they arrive without intervention.